Automated and Intelligent generation of Massing Models and Visualisation from Design Brief

Team 7

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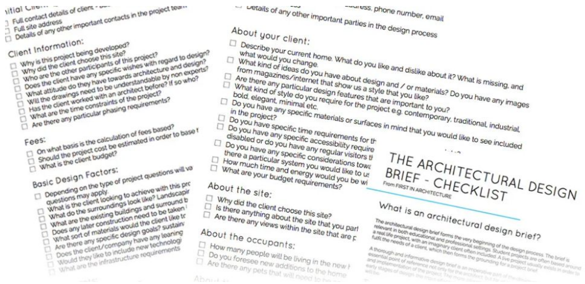
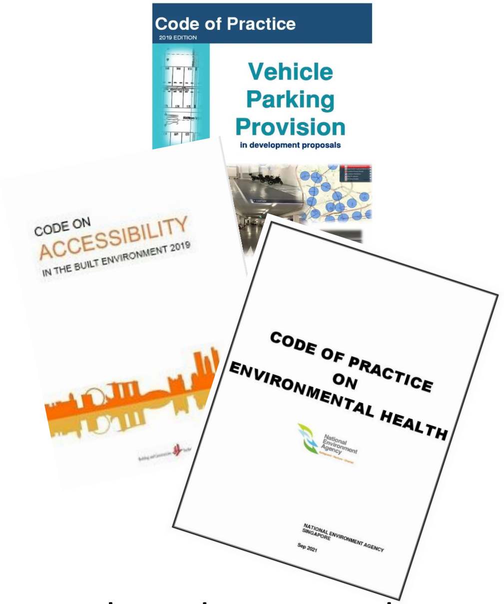
The routine of understanding design briefs



❶ Obtain briefs & info from client

❷ Architects extract info and convert

it into usable data through tables



❸ Apply and extract relevant parameters

❹ Generate massing models

Problems

- Briefs provided in different formats
- Lengthy pieces of text
- Varying degrees of fixed and variable info and parameters

<u>Routine and mundane task that takes time</u>

Solutions

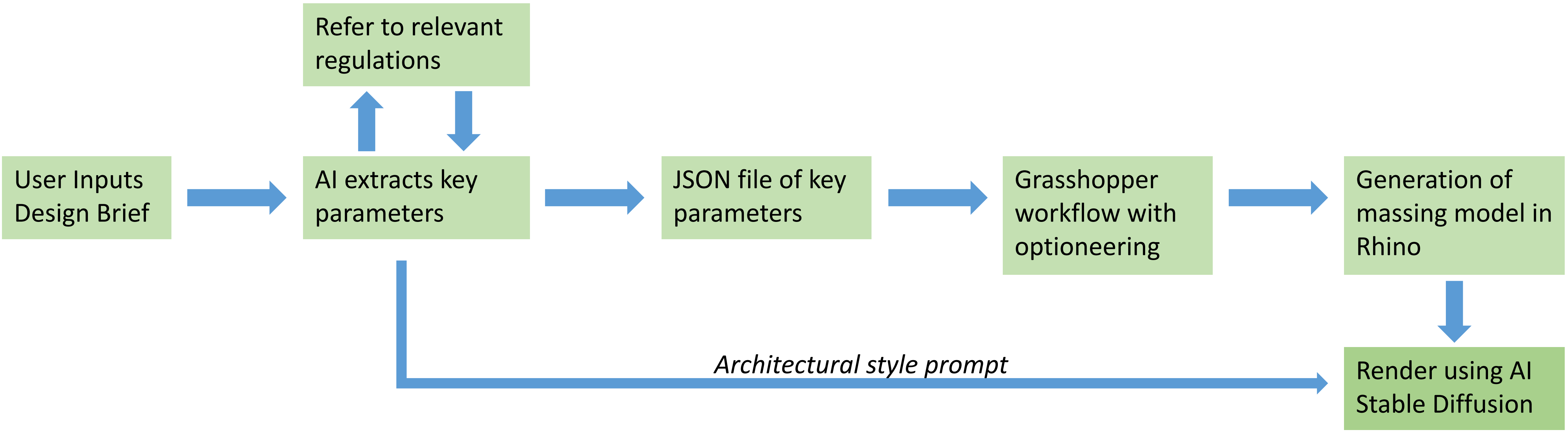
- Automation of steps 1 to 4 mentioned earlier

Use AI to pick key data from design brief → Apply relevant regulatory and planning parameters → Pass info to parametric modelling platform → Generate quick massing model → Quick render of model based on preferred style

Objectives

- Allow architects and project members quickly get overview of project and saves time
- Assist them to determine scope of works & resource planning immediately at the onset of receiving brief
- Allow more accurate quoting of fees and high level cost estimation before project begins
- Greatly useful for feasibility studies and design competitions

Crafting the solution



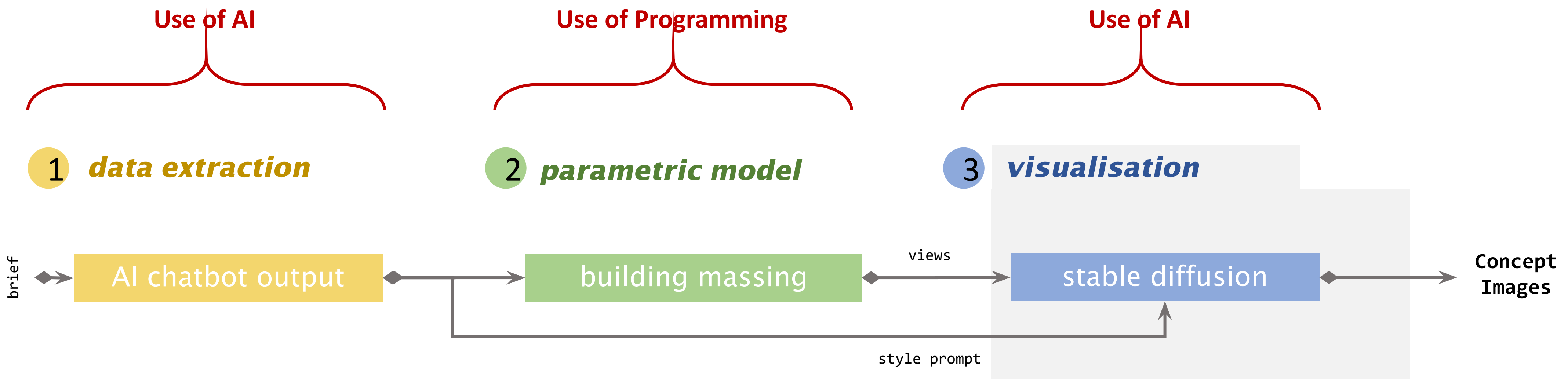
Tools used

- Botpress – A ChatGPT chatbot powered by the latest in LLMs and GPT by OpenAI
- Rhino+Grasshopper – Rhino for 3D modelling; Grasshopper for parametric visual programming
- Stable Diffusion – A deep learning, text-to-image model based on diffusion techniques

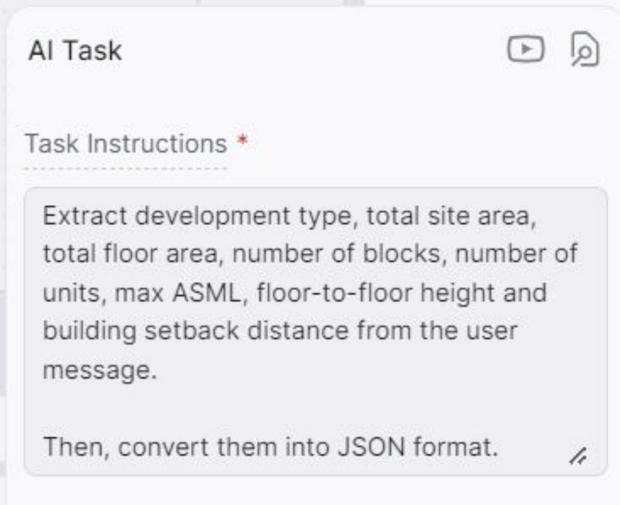
Inputs required

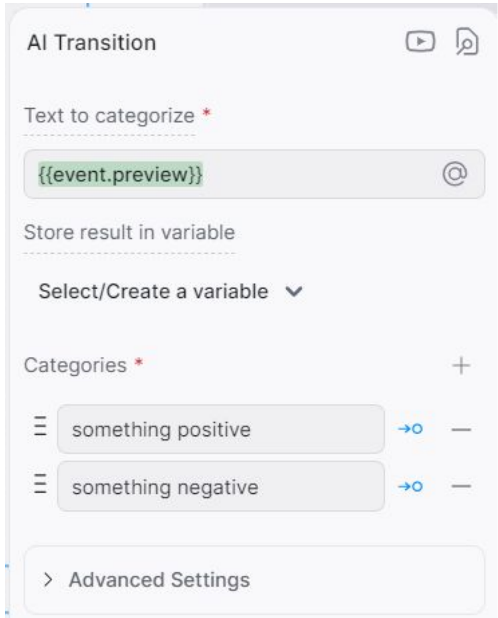
- Design Brief
- Site boundary in closed planar polyline

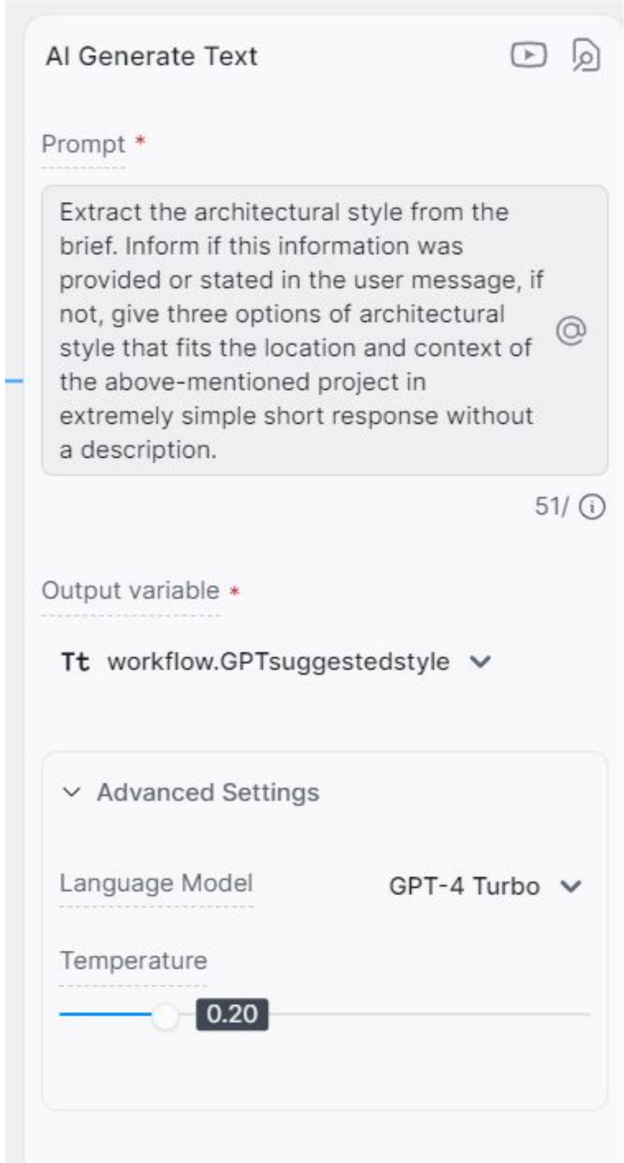
3 Step solution

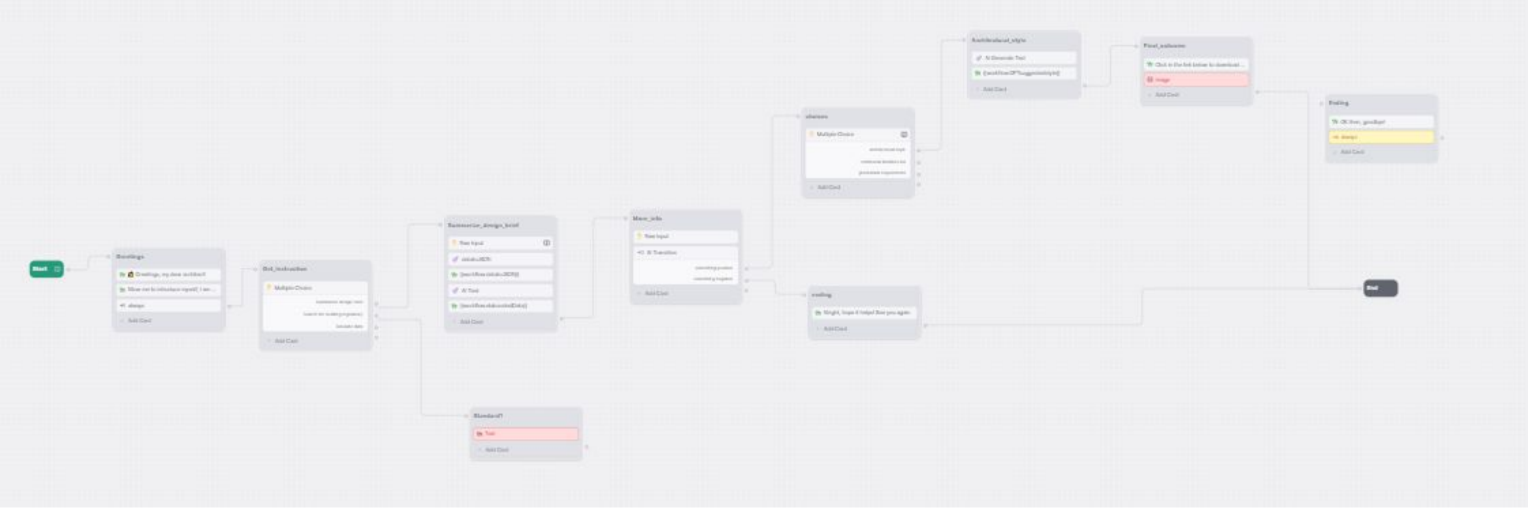


1. Use AI chatbot to extract key parameters from design brief









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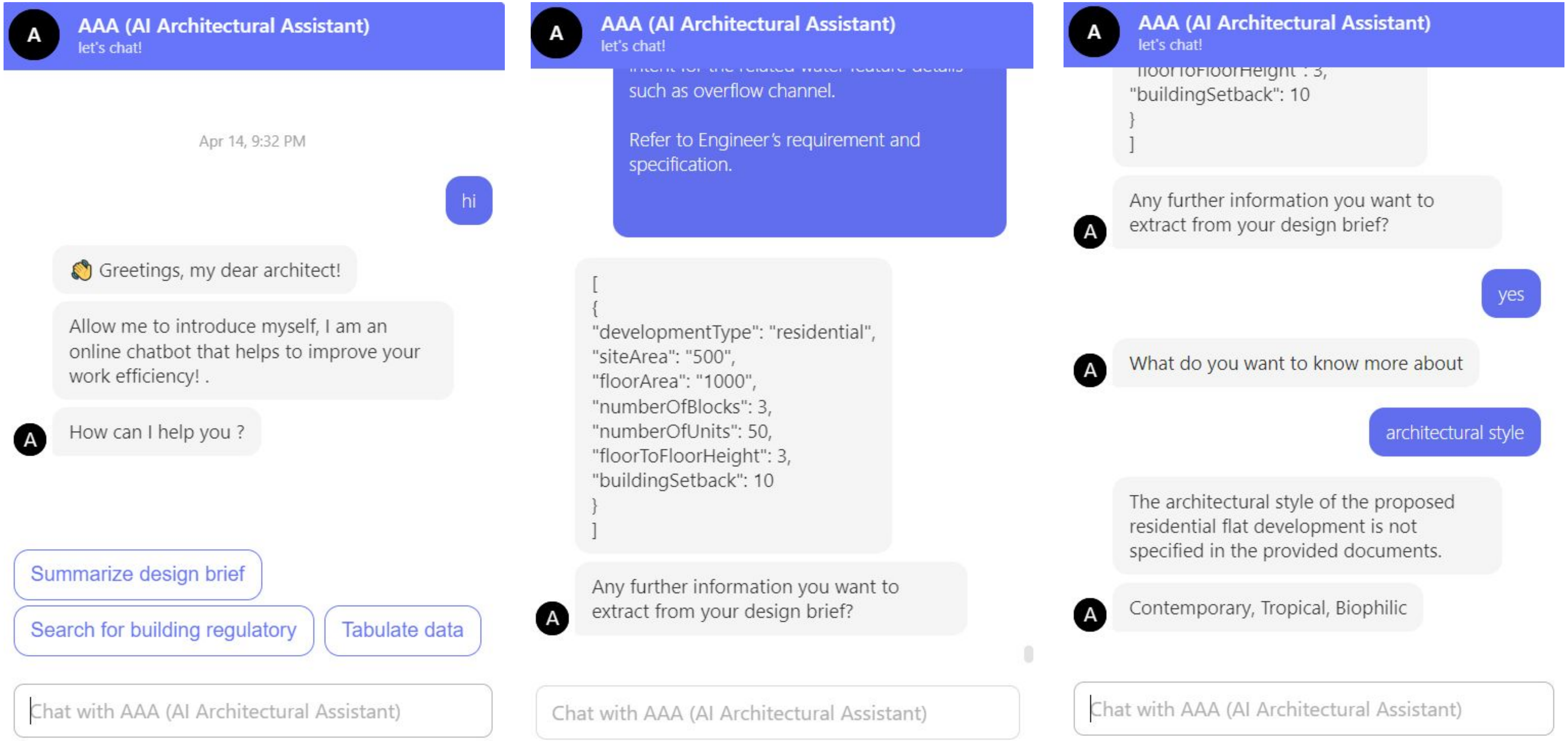
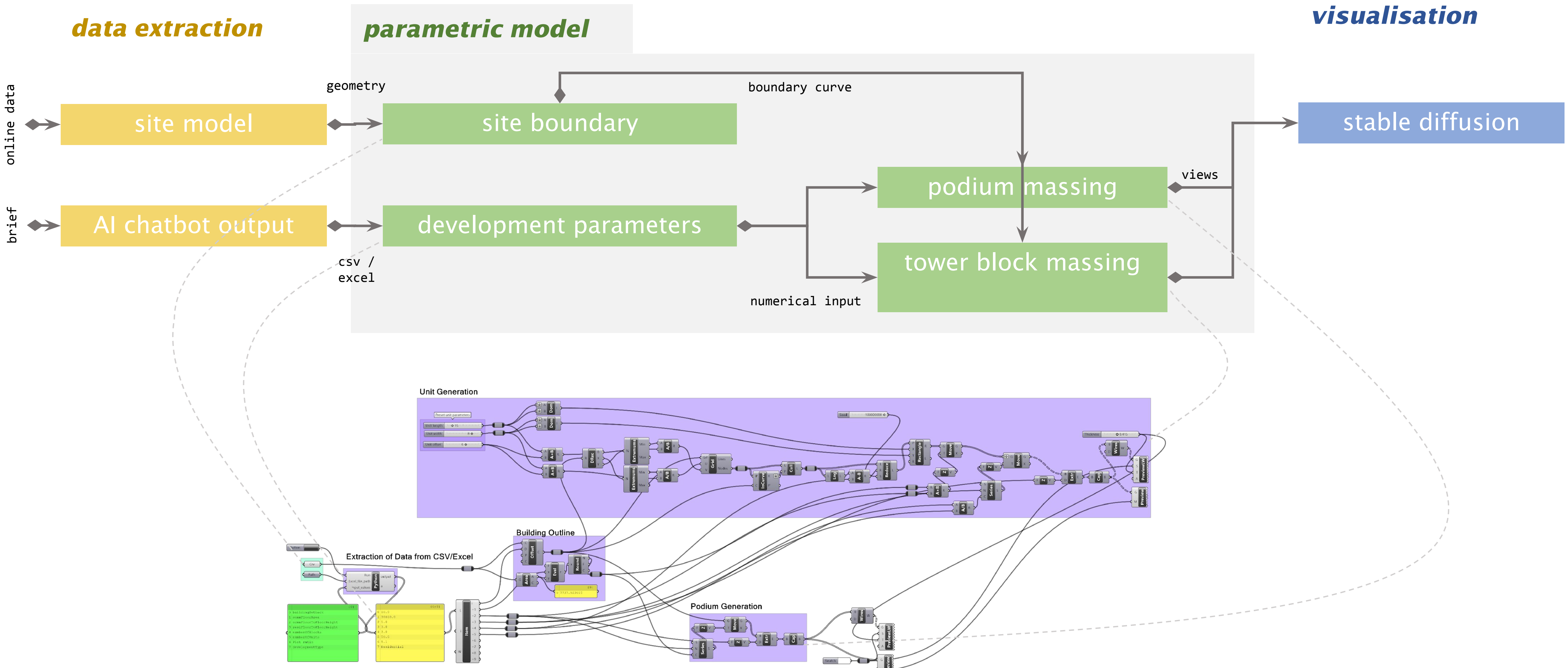


2. Generate massing model in Rhino+Grasshopper with extracted parameters



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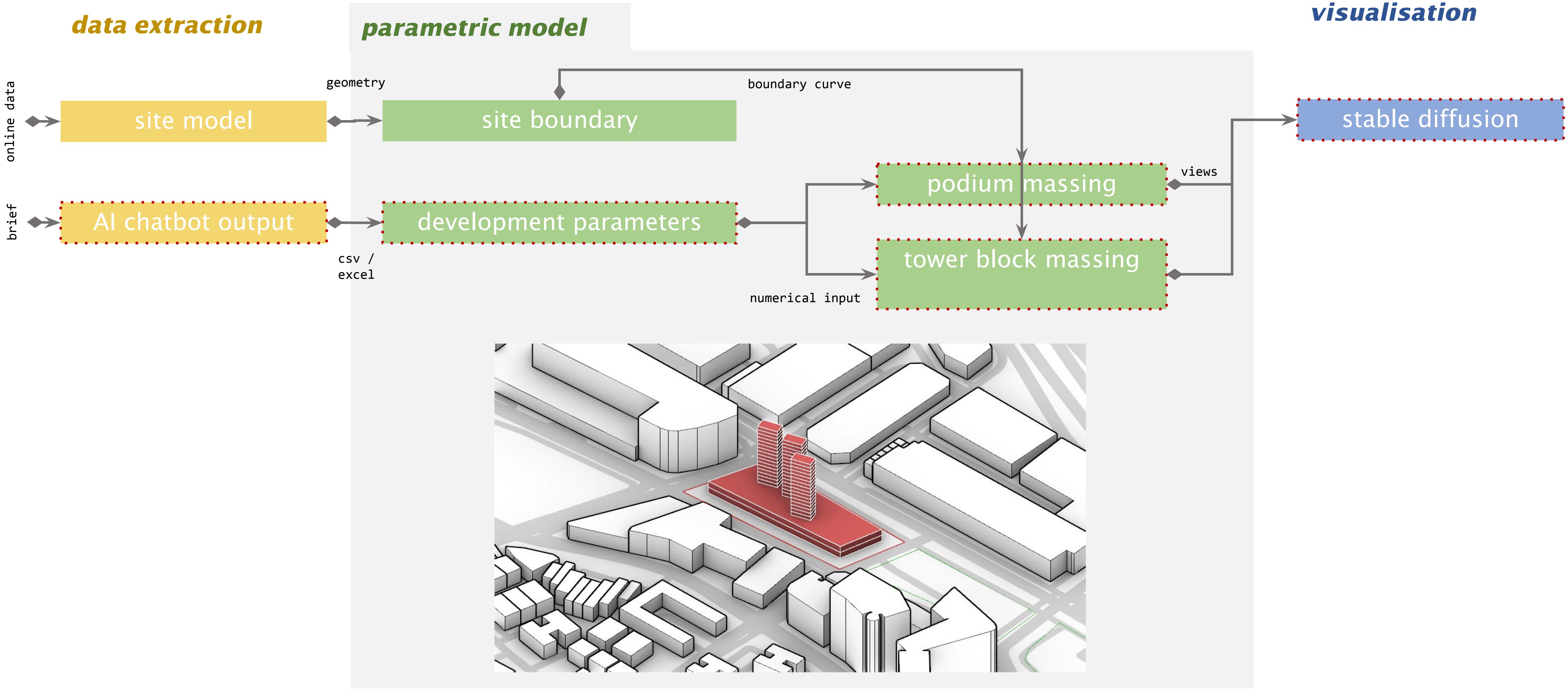
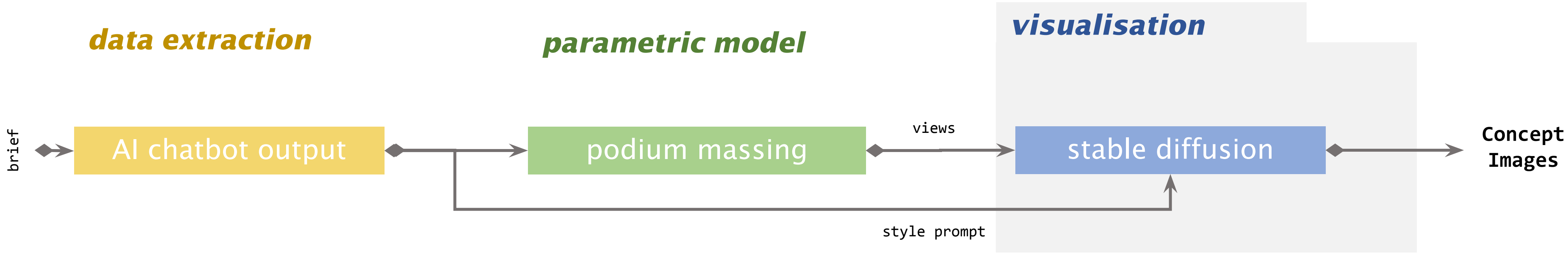


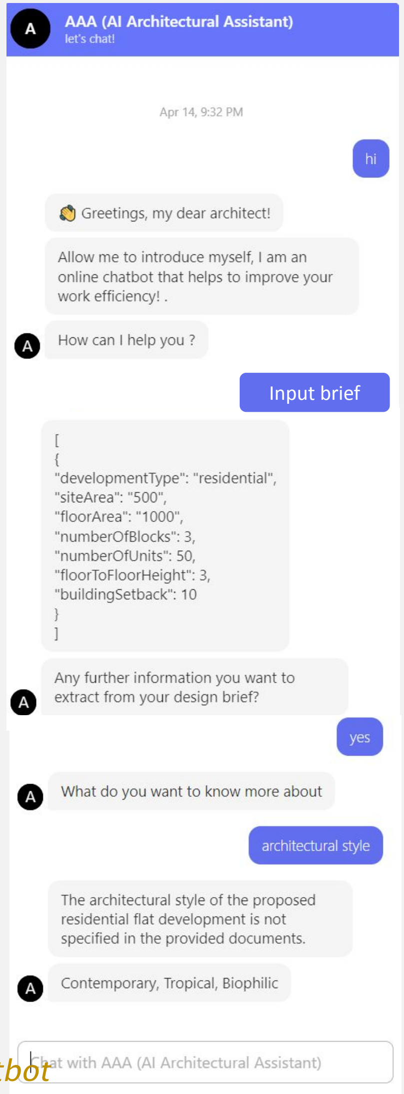
3. Render in Stable Diffusion based on styles extracted from design brief



From Brief to Pitch in 15 mins

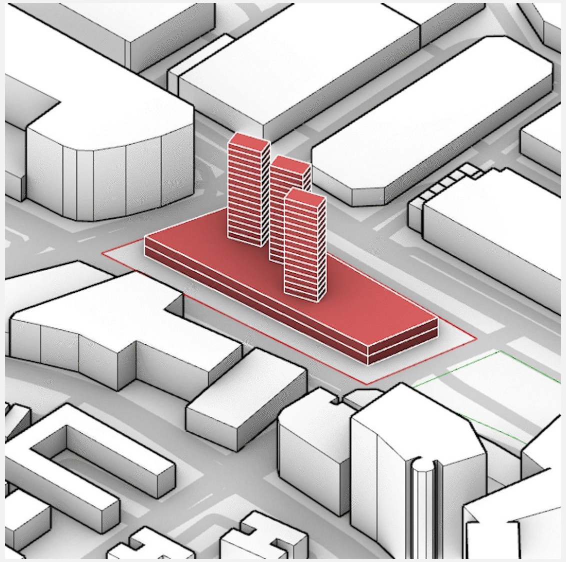
BRIEF

data extraction



AI chatbot

parametric model



Design tool

visualisation

Stable diffusion

PITCH

Starting small and simple to making it more comprehensive

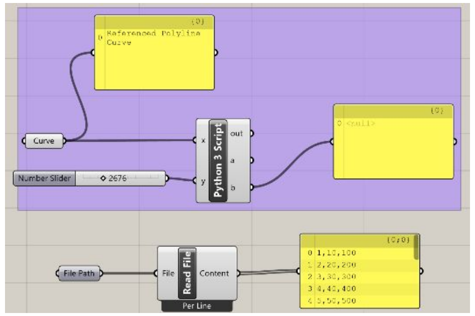
1

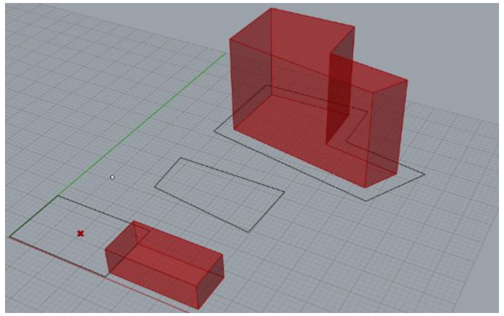
Start with simple prompts

Create chatbots

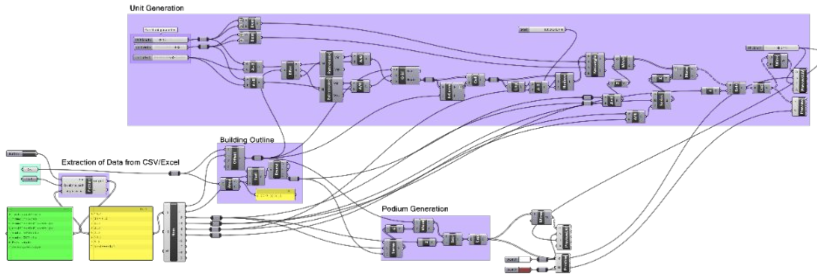
2

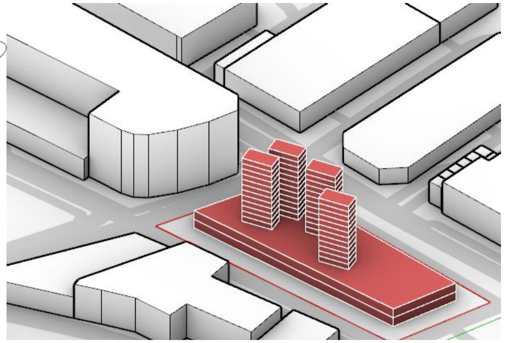
Model generation with simple parameters





Model generation with multiple and complex parameters





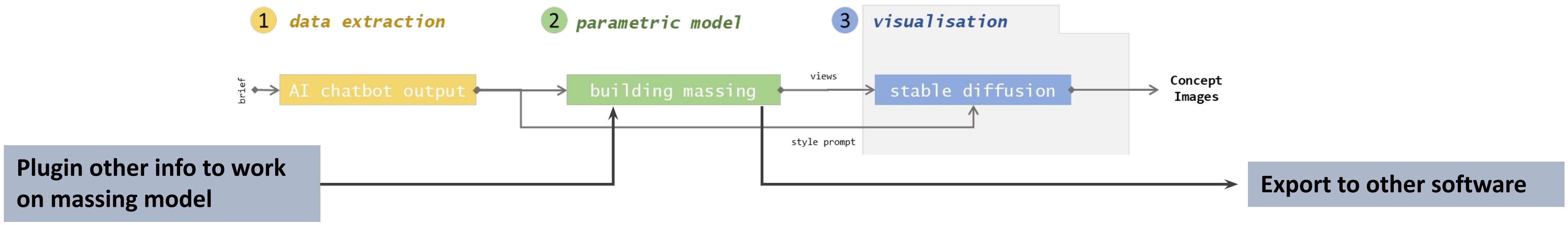
3

Simple renders

Photorealistic renders with ControlNet

Next steps to Scale up

1. **Unified interface** □ Complete solution within 1 app/ program
2. Extracting **more granular design parameters** – colours, signage etc
3. Application of more **detailed regulatory requirements** e.g. corridor widths, escape staircase determination etc
4. Expand optioneering – **AI data analysis** to select most efficient or most preferred layout based on client's requirements
5. Expand jurisdiction beyond Singapore – use AI to analyse and apply regulatory requirements for **projects overseas**
6. Picking sites directly from **Google Earth**
7. **Link final solutions to BIM softwares** such as Revit and Archicad with IFC-SG information incorporated for further design development
8. **Plugin scripts** that can work on top of generated massing model such as façade generation or parking layout generation
9. **Iterative development** with multiple stakeholders and interoperability with multiple programs to allow solution to be used from beginning to end of entire project cycles



Thank you ☺️

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